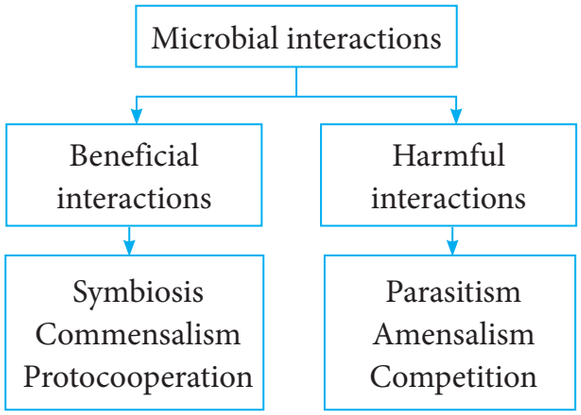
harmful relationships (Flowchart 10.1). Some of the interaction and interrelationship have been discussed in this connection in Table 10.2.



Flowchart 10.1: Microbial interactions

10.5.1 Beneficial Interactions

The beneficial interactions such as symbiosis (mutualism) and commensalism are found to operate among the soil inhabitants.

Symbiosis (mutualism)

Mutualism is an example of symbiotic relationship in which each organisms benefits from the association. One type of mutualistic association is involving the exchange of nutrients, between two species, a phenomenon called syntrophism. Many microorganisms synthesise vitamins and amino acids in excess of their nutritional requirements. Other have a requirement for one or more of these nutrients. Symbiosis is an obligatory relationship between two populations that benefit both the population. Both populations live together for mutual benefit.

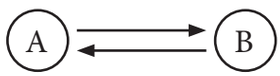


Table 10.2: Types of microbial interaction in soil

Interaction	Microorganisms A	Microorganisms B
Neutralism	No effect	No effect
Commensalism	+	No effect
Amensalism	No effect	-
Mutualism Synergism Protoco-operation Symbiosis	+	+
Competition	-	-
Parasitism	+	-
Predation	+	-
+ = positive effect – = negative effect		

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